

Comparative Analysis of Usability, Performance, and Security of Open-Source, Windows-Based Password Manager Applications Based on ISO/IEC 25010

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Abstract—Many workers need multiple applications to carry out their works. The number of applications makes users have to set several accounts and passwords for various application services. This creates complexity in managing it manually so that users switch to using password manager applications, both free and paid. KeePass, KeePassXC, and Password Safe are three examples of open-source password manager applications developed as low-cost solution. In general, open-source password manager applications have disadvantages compared to similar paid applications. Analysis of open-source password manager applications is necessary to find out their specifications, advantages, and disadvantages so that they can be taken into consideration by users in choosing a suitable password manager application. In this study, a comparative analysis of three open-source, Windows-based password manager applications i.e., KeePass, KeePassXC, and Password Safe was carried out using the Usability, Performance, and Security parameters based on ISO/IEC 25010. The result of this study indicates that on the parameter of Usability, the KeePassXC excels in all six categories. In parameter of Performance, the Password Safe excels in one category, the KeePass also excels in one category, and the KeePassXC and Password Safe are equivalent in one category. Furthermore, on parameter of Security, KeePass outperformed others in one category and for the rest four categories, all applications are equivalent.

Keywords— *comparative analysis, open-source, password manager, performance, security, usability*

I. INTRODUCTION

Internet users reached 150 million people in Indonesia in 2019 with 56% penetration spread throughout the region [1]. Utilization of Internet services is spread in several fields such as communication, education, social media, e-commerce, e banking and recreation [2]. Such many uses of Internet services cause users to have and set up multiple accounts and passwords for each different service.

An increase in the number of accounts and passwords held by individuals can lead to poor password management such as using the same password on different services, passwords that are common and easy to guess, passwords that are not complex, or passwords that are not changed regularly. . Not only can it harm individuals, the habit of using bad passwords can also cause harm to the company when it is done by employees. The risk is that it can cause sensitive data leakage which can result in loss of business and reputational damage.

Password manager applications can be an alternative to overcome the above-mentioned problems and can be a means of implementing good password management practices [3].

Password manager application is software that helps users in creating, storing, and managing passwords. Applications can use local databases or use a server or cloud to store passwords. Passwords are stored in encrypted form to ensure an adequate level of protection. The user only needs to memorize one very strong password as the master key, which gives the user access to the entire password database [3]. In addition, many password manager applications are featured with additional functions such as web browser integration and multi-factor authentication [4]. There are password manager applications that are openly available and developed as low-cost solution to good password management practices. The open-source password manager applications provide an opportunity for its users to use, view the code of, and develop the application for free [4].

Research on open source password manager applications needs to be done to find out the specifications, advantages and disadvantages as a consideration in using the application. Comparative study can be chosen for this purpose since the results of comparative research are in the form of an explanation of the differences and similarities of the research object. One research method that can be used is an exploratory study because it is an approach to finding information about a system [5].

In this study, a comparative analysis based on Usability, Performance, and Security was carried out on three open-source password management applications, namely KeePass, KeePassXC and Password Safe, based on ISO/IEC 25010. These three applications were chosen because they are open-source password management applications that store passwords locally and offline, making it easy to assess at a relatively low cost. The definition of benchmarking parameters for Usability, Performance, and Security of applications in this study was carried out based on ISO/IEC 25010. It is an international standard that can be used by independent assessors and provides consistent terminology for determining, measuring, and evaluating the quality of software products [6]. The comparison parameters chosen for the comparative analysis are Usability, Performance and Security of the application because Usability relates to the effectiveness of use, Performance relates to the speed of application processing and Security relates to the application mechanism in protecting the privacy of user data. In addition, the comparison parameters were chosen because Usability, Performance, and Security have the most significant influence on the quality in use for the main users, those who interact with the application to achieve the main goal [6]. The expected result of this research is a comparison of the Usability, Performance, and Security of three open source password

management applications, namely KeePass, KeePassXC, and Password Safe. The comparison is expected to be a reference and consideration in the use of the application.

II. RESEARCH METHOD

In this research, we used exploratory study to determine the characteristics of three password manager applications. The stages of this research correspond to consisting of the process of collecting, analyzing, and interpreting observations about known designs, systems, or models, or about abstract theories or subjects [5]. The next section will explain the stages of research based on exploratory study methods consisting of research preparation, data collection and comparative analysis.

A. Research Preparation

The research preparation stage contains activities aimed at preparing and ensuring all the needs needed for research are met. In the preparation stage of this research, the following activities were carried out:

- Searching and studying literature that discusses passwords, password manager applications, Usability, Performance and Security of applications based on ISO/IEC 25010, as well as comparative analysis. The purpose of this activity is to gain knowledge that supports this research.
- Studying the documentation of the application under study. The purpose of this activity is to obtain information on the specifications of the application under study.
- Preparing the laptop used to do this research.
- Preparing applications to be researched, namely KeePass 2.4.7, KeePassXC 2.6.4 and Password Safe 3.55 for the Windows platform desktop version.
- Preparing the applications used in this research, namely VirtualBox Linux 64-bit 6.18, Process Monitor 3.61, SysGauge Windows 64-bit 7.6.38 and Windows 10 64-bit Pro operating system.
- Creating three units of virtual machines in VirtualBox that have identical specifications as the working environment for each application under study. This activity is carried out by installing Windows 10 Pro 64-bit on one virtual machine and then cloning three virtual machines.
- Installing the Process Monitor and SysGauge applications on each virtual machine of the work environment.
- Installing the KeePass application on Virtual Machine 1, KeePassXC on Virtual Machine 2 and Password Safe on Virtual Machine 3.
- Conducting trials to ensure these applications can work well so that they are ready to be used for research.

B. Data Collection

1) Data collection on parameter of Usability

Data collection on parameter of Usability used a survey method by distributing questionnaires to at least 20 respondents to obtain statistically significant description [7]. We used purposive sampling technique in selecting

respondents based on the criteria required by the researcher. The criteria are cadets/intermediate cadets of the Politeknik Siber dan Sandi Negara who were willing to carry out the tasks that have been determined by the researcher. These tasks were determined based on the things that must be done by the main user or people who interact with the application to achieve the main purpose of the application, namely storing and managing passwords [6]. These tasks are as follows:

a) *Install* KeePass, KeePassXC, and Password Safe applications on Windows operating system.

b) *Create a new database* on the KeePass, KeePassXC, and Password Safe.

c) *Create a new group* in the KeePass, KeePassXC, and Password Safe.

d) *Add a new entry* in the KeePass, KeePassXC, and Password Safe.

e) *Create a password* on the KeePass, KeePassXC, and Password Safe.

The questionnaire distributed to respondents has options using a Likert scale. The Likert scale is designed to convince respondents to answer at various levels on each question item contained in the questionnaire [8]. The Likert scale on the questionnaire is based on the Usability parameter in ISO/IEC 25010, which has six categories, namely Recognition of Conformity, Learnability, Operability, User Error Protection, User Interface Aesthetics, and Accessibility. The Likert scale contains statements that are arranged based on the definition of each of these categories in ISO/IEC 25010. This Likert scale will show the usability of the KeePass, KeePassXC and Password Safe applications based on the respondent's level of agreement and as a measurement scale in the Usability parameter testing instrument that has a value scale 1-5 as in Table I. After the survey results are obtained, the percentage of respondents' acknowledgment level (RAL) is calculated for the assessment of the usability of the application. Calculation of the RAL uses the Equation (1) [9].

$$RAL (\%) = \frac{\text{total scores}}{\text{total ideal score}} \times 100\% \quad (1)$$

TABLE I. CRITERIA OF ASSESSMENT

Criteria	Symbol	Score
Strongly Disagree	SD	1
Disagree	D	2
Fair	F	3
Agree	A	4
Strongly Agree	SA	5

After the results of the percentage of RAL are obtained, the results are compared with the range of respondent's acknowledgment as in Table II.

TABLE II. RANGE OF RESPONDENTS' ACKNOWLEDGMENT

Range (%)	Criteria
0% - 20%	Strongly Disagree
21% - 40%	Disagree
41% - 60%	Fair
61% - 80%	Agree

Range (%)	Criteria
81% - 100%	Strongly Agree

2) Data collection on parameter of Performance

Data collection on parameter of Performance was done by simulating and monitoring application performance using Process Monitor and SysGauge applications [10]. Parameter of Performance, according to ISO/IEC 25010, has three categories, namely Time Behavior, Resource Utilization, and Capacity. In detail, data collection from each category on Performance parameter in ISO/IEC 25010 is described as follows:

a) *Time Behavior category data collection* is done by running the application and then trying to save the password. When saving passwords, the Process Monitor application is run using the Event Profiling filter to monitor the time it takes the application to store passwords. The Profiling Event filter is used to scan all active threads in the system to generate a profile and record user time (user time) and user CPU kernel time (kernel time) consumed by the active thread [11]. User time (user time) is the amount of time the processor works on a particular application program and kernel time (kernel time) is the amount of time the processor works on operating system functions that are connected to certain application programs [12]. The monitoring results are then saved in .csv format for further comparative analysis. Password storage and monitoring experiments using the Process Monitor application were performed 100 times manually using the same password in each trial. The fields that are filled in during the password storage experiment are Title, Username and Password.

b) *Data collection for the Resource Utilization category* is done by running the application and then trying to save the password. When saving a password, the SysGauge application is run to monitor the CPU and memory (RAM) usage of the computer that the application needs to save the password. The monitoring results are then saved in pdf format for further comparative analysis. Password saving and monitoring experiments using the SysGauge application were performed 100 times manually using the same password in each trial. The fields that are filled in during the password storage experiment are Title, Username and Password. Furthermore, the search for information about the size of the installer file and the directory of the application program files under study was also carried out.

c) *Data collection for the Capacity category* is carried out by measuring the size of the password database after storing passwords 100 times manually using the same password in each trial.

3) Data collection on parameter of Security

Data collection on parameter of Security was done by looking for information about application security based on Security parameter in ISO/IEC 25010 [10] which has five categories, namely Confidentiality, Integrity, Non-repudiation, Accountability, and Authenticity. In detail, the data collection on parameter of Security is described as follows:

a) *Confidentiality category data collection* was done by looking for information on the type of encryption algorithm used by the application under study.

b) *Integrity category data collection* was done by checking whether the application under study is able to detect any modifications to the data stored in the application.

c) *Non-repudiation category data collection* was carried out by checking whether the application under study is able to prove that an event or action has occurred in the application. This is done by checking the existence of the system log in the application.

d) *Accountability category data collection* was done by checking whether the application under study is able to uniquely trace the actions of an entity. This is done by checking the condition of the system log on the application.

e) *Authenticity category data collection* was done by checking whether the identity of a subject or resource can be proven as claimed by the application under study. This aimed at checking whether there is a system that can be used to prove a subject or resource in the form of a password database as claimed by the application under study.

III. RESULT AND DISCUSSION

After the research preparation and data collection stages are carried out, the following step is the explanation of the results of a comparative analysis of the Usability, Performance, and Security parameters of the KeePass, KeePassXC and Password Safe applications.

A. Comparison of Usability

Usability benchmarking parameter consists of the categories Acknowledgment of Suitability, Learnability, Operability, User Error Protection, User Interface Aesthetics and Accessibility. In this section, a comparative analysis of the results of data collection is carried out for the comparison parameter of usability in the form of the percentage of respondents' approval level on the usability of the application under study. In Table III, the results of the comparison of the parameters of the application under study are presented.

The analysis and interpretation of the Usability parameter was carried out with the categories of Suitability Acknowledgment, Learnability, Operability, User Error Protection, Graphical User Interface (GUI) Aesthetics, and Accessibility. The results obtained are that the KeePassXC application has an average respondent approval level of 78.11% for all Usability categories, greater than the KeePass application which has an average respondent approval level of 74.99% and the Password Safe application which has an average respondent's approval level is 72.20%.

TABLE III. COMPARISON OF USABILITY

Category	Application		
	KeePass	KeePassXC	Password Safe
Suitability acknowledgment	76,77%	79,67%	73,87%
Learnability	74,83%	78,70%	71,29%
Operability	77,74%	78,70%	72,90%
User's error protection	74,19%	76,45%	70,32%
GI aesthetic	69,35%	77,09%	72,25%

Category	Application		
	KeePass	KeePassXC	Password Safe
Accessibility	77,09%	78,06%	72,58%
Average	74,99%	78,11%	72,20%

B. Comparison of Performance

Performance comparison parameters consist of the categories of Time Behavior, Resource Utilization and Capacity. In this section, a comparative analysis of the results of data collection is carried out for performance comparison parameters in the form of time required to store passwords, work environment resources used when storing passwords and capacity of application password databases.

1) Time Behavior

In this section, a comparative analysis of the average duration of the password storage process for the application under study is carried out. In Table IV, a comparison of the average data duration of the application password storage process is presented in the form of user time and kernel time. For a clear comparison of the time behaviour between the three applications, we can see in Fig. 1.

Furthermore, the analysis and interpretation of the Performance parameters with the category of Time Behavior is carried out. The results obtained are that the Password Safe application has an average user time of 1.147 seconds and a kernel time of 1.819 seconds, which is smaller than the KeePass application which has an average user time of 13.867 and a kernel time of 5.998 seconds and the KeePassXC application which has an average user time of 15.599 and a kernel time of 4.020. second.

TABLE IV. COMPARISON OF TIME BEHAVIOR

Time Duration	Application		
	KeePass	KeePassXC	Password Safe
User Time	13,867 s	15,599 s	1,147 s
Kernel Time	5,998 s	4,020 s	1,819 s

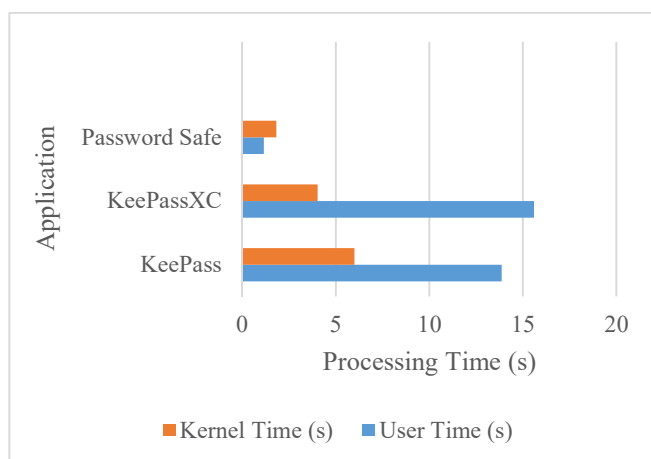


Fig. 1. Comparison of time behavior

2) Resource Utilization

In this section, a comparative analysis of the average data utilization of the application resources under study was carried out. In Table V, a comparison of the average data utilization

of the application resources under study is presented. For a clear comparison of the capacity between the three applications, we can see in Fig. 2. In general, memory usage is more than CPU, Installer files, and directory files. It can be clearly seen that KeePassXC uses more resources than the other two applications.

TABLE V. COMPARISON OF RESOURCE UTILIZATION

Component	Application		
	KeePass	KeePassXC	Password Safe
Central Processing Unit (CPU)	1,92 %	0,72 %	0,56 %
Memory	51,218 MB	88,306 MB	75,426 MB
Installer file size	4,10 MB	42,9 MB	16,7 MB
Directory size	8,43 MB	110 MB	14,7 MB

Furthermore, analysis and interpretation of performance parameters is carried out with the category of Resource Utilization. The results obtained are the KeePass application uses the CPU with an average percentage of 1.92%, uses memory with an average of 51.218 MB, has an installer file size of 4.10 MB and a file directory size of 8.43 MB. The KeePass application is more resource efficient than the Password Safe application which uses CPU with an average percentage of 0.56%, uses memory with an average of 75.426 MB, has an installer file size of 16.7 MB and a file directory size of 14.7 MB. The KeePass application is also more resource efficient than the KeePassXC application which uses CPU with an average percentage of 0.72%, uses memory with an average of 88.306 MB, has an installer file size of 42.9 MB and a file directory size of 110 MB.

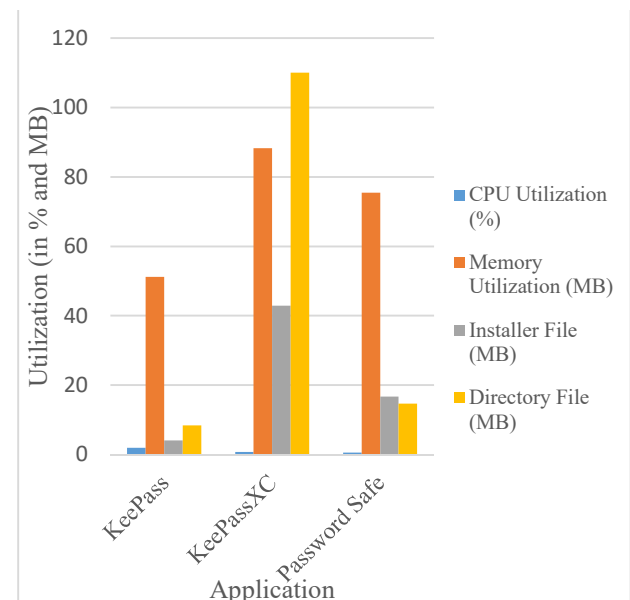


Fig. 2. Comparison of resource utilization

3) Capacity

In this section, a comparative analysis of the application capacity data is carried out. In Table VI, a comparison of the application capacity data is presented. For a clear comparison

of the capacity between the three applications, we can see in Fig. 3.

TABLE VI. COMPARISON OF CAPACITY

Capacity	Application		
	KeePass	KeePassXC	Password Safe
Password database size	476 KB	164 KB	164 KB

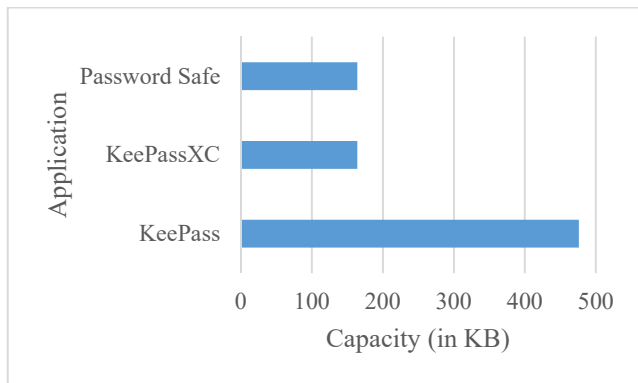


Fig. 3. Comparison of capacity

Furthermore, the analysis and interpretation of the Performance parameters with the category of Capacity is carried out. The results obtained after saving 100 passwords are that the KeePassXC and Password Safe applications have a database size of 164 KB, smaller than the KeePass application which has a database size of 476 KB. This shows that the KeePassXC and Password Safe applications have a larger password database capacity than the KeePass applications.

C. Comparison of Security

The Security Parameter consist of five categories namely Confidentiality, Integrity, Non-Repudiation, Accountability, and Authenticity. In Table VII, a comparison of data on the security parameter of the three applications studied is presented.

TABLE VII. COMPARISON OF SECURITY PARAMETERS

Category	Application		
	KeePass	KeePassXC	Password Safe
Confiden- tiality	AES/ Rijndael 256- bit, ChaCha20 256-bit, Twofish, Serpent, GOST	AES/ Rijndael 256- bit, Twofish, ChaCha20 256- bit	Twofish 256-bit
Integrity	Data collection was not carried out because all data related to passwords was stored in an encrypted password database file and not stored in the application.	Data collection was not carried out because all data related to passwords was stored in an encrypted password database file and not stored in the application.	Data collection was not carried out because all data related to passwords was stored in an encrypted password database file and not stored in the application.
Non- repudiation	Has no system log	Has no system log	Has no system log

Category	Application		
	KeePass	KeePassXC	Password Safe
Account- ability	Data collection was not carried out due to the absence of system log	Data collection was not carried out due to the absence of system log	Data collection was not carried out due to the absence of system log
Authenti- city	The application proves the identity of the subject or resource in the form of a password database using a master key called Master Password	The application proves the identity of the subject or resource in the form of a password database using a master key called Master Password	The application proves the identity of the subject or resource in the form of a password database using a master key called Master Password

Next, the analysis and interpretation of the Security parameters is carried out with the categories of Confidentiality, Integrity, No Denial, Accountability and Authenticity.

In the Confidentiality category, the results obtained are that KeePass is supported by encryption algorithms AES/Rijndael 256 bit, ChaCha20 256 bit, Twofish, Serpent, and GOST. Compared to KeePass, KeePassXC has less algorithm support, namely AES/Rijndael 256 bit, Twofish 256 bit and ChaCha20 256 bit. Password Safe has the least encryption algorithm which is only Twofish 256 bit. In the Integrity category, data collection was not carried out on the three applications because all data related to passwords was stored in an encrypted password database file and not stored in the application.

In the Non-Repudiation category, the result obtained is that the three applications, namely KeePass, KeePassXC, and Password Safe, do not have a system log. In the Accountability category, data collection was also not carried out because the three applications did not have a log system.

And lastly, in the Authenticity category, the results obtained is that the three applications can prove the identity of a subject or resource in the form of a password database using master key information which is referred to by several different terms in each application under study, namely Master Password (on KeePass), Password (KeePassXC), and Safe Combination (Password Safe).

IV. CONCLUSION

In this study, a comparative analysis of Usability, Performance, and Security has been carried out on three open-source password management applications, namely KeePass, KeePassXC, and Password Safe, based on the Windows Desktop platform.

Based on the parameter of Usability with categories including Acknowledgment of Suitability, Learnability, Operability, User Error Protection, User Interface Aesthetics, and Accessibility, KeePassXC has a usability with an average respondent approval level greater than KeePass and Password Safe.

Based on the parameter of Performance category of Time Behavior, Password Safe is faster in processing than KeePass and KeePassXC. In the category of Resource Utilization, KeePass is more efficient in the use of work environment

resources than KeePassXC and Password Safe. In the category of Capacity, KeePassXC and Password Safe have a larger password database capacity than KeePass.

Based on the parameter of Security with the category of Confidentiality, KeePass application has more choice of encryption algorithms compared to KeePassXC and Password Safe. In the categories of Integrity, Non-Repudiation, Accountability, and Authenticity, the three applications are equivalent where all data related to passwords is stored in an encrypted password database file and is not stored in the application system log and can also prove the identity of a subject or resource using master key information whose names are different for each application.

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